

TABLE 3
CONNECTION AMONG ES32 PIN, BT66, BC547, SPEAKER, BT1 AND RESISTOR

ESP32	BT66	Resistor R1	BC547	Speaker	+5V	GND	BT1(-)	BT1(+)
GND	Pin1		E			GND	-	
GPIO15	Pin2							
5V					+5V			
	Pin3	R1						
		R1	B					
			C	2				
				1(+)				+

Fig. 4 shows the connection diagram of the notifier on a breadboard. It will help in understanding how to connect components on the breadboard.

Table 3 shows connections among ES32 pins, BT66, BC547, speaker, 9V battery (BT1), and the resistor.

BLE is widely considered for wireless indoor localisation systems (ILS) in which BLE-received signal strength (RSS) is most often used to derive the location of the target. The setup was tested in building premises with multiple adjacent rooms, separated by walls, windows, and doors. It worked successfully up to 15-18 metres, including opaque separations in a non-line of sight (NLOS) condition.

On a conclusive note, BLE detects individual persons, provided the MAC address on the respective

smartwatch can be identified. To ensure gender equality, the code was tested with the father's smartwatch also. And it worked!

In the code, configure the MAC address with the MAC address of your smartwatch, then select the board as ESP32 and upload the code to the ESP32 board. Fig. 5 shows the snippet of the source code.

Construction and testing

After uploading the source code, assemble the ESP32 board on a breadboard along with other components as shown in Fig. 3. Now power the device and keep it in the home. Next, go out wearing your smartwatch, and when you come near the home, the device will be able to sense it. It starts notifying using the speaker.

The same device can be changed with a relay to automatically turn on the lights, AC, and door, when you come home. It can also be modified as a security system. When you are at home, the device turns on; otherwise, it automatically locks the home. Overall, the notifier can be modified for many use cases according to your creativity. **EFY**

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EFY Note

The source code of this project is available for free download at www.electronicsforu.com



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